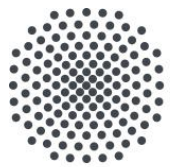


A Multi-Domain Quantum Low-Code Platform



University of Stuttgart

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Institute for Architecture of Application Systems

Motivation

- Quantum computing could generate \$620 billion up to \$1.27 trillion USD in economic value by 2035 across industries^[1]



Finance

Portfolio optimization
Risk analysis
Derivatives pricing



Logistics

Vehicle routing
Supply chain
Scheduling



Chemistry

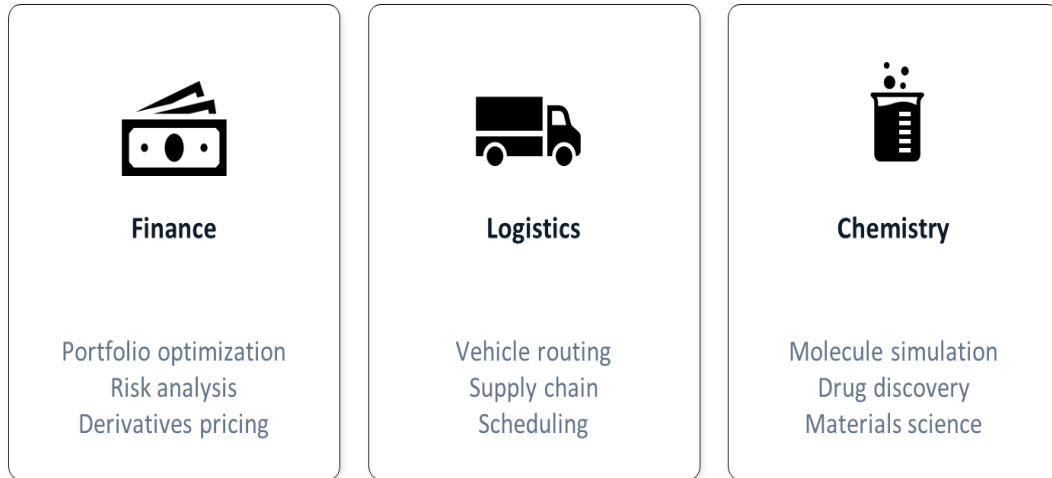
Molecule simulation
Drug discovery
Materials science

[1] McKinsey & Company. (2023). Quantum Technology Monitor: April 2023.

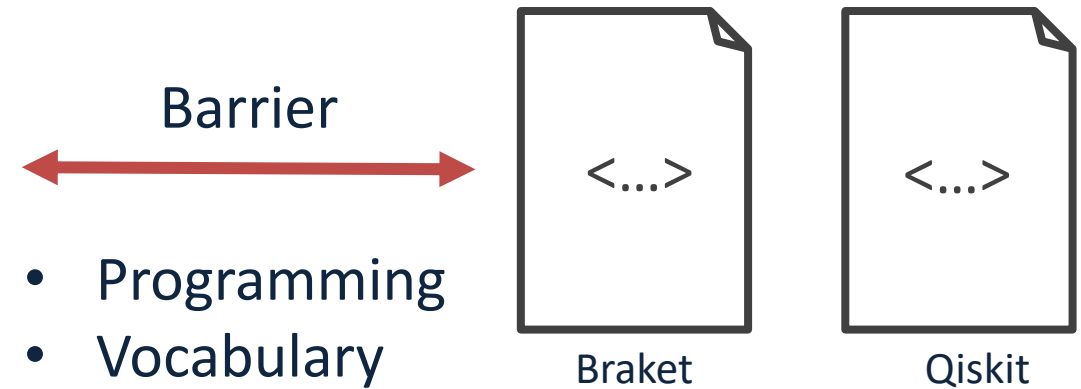
<https://www.mckinsey.org/~media/mckinsey/business%20functions/mckinsey%20digital/our%20insights/quantum%20technology%20sees%20record%20investments%20progress%20on%20talent%20gap/quantum-technology-monitor-april-2023.pdf>

Motivation

„**What** needs to be solved?“



„**How** to solve a problem?“



Existing Approaches

- Quantum SDKs
 - Require programming and quantum expertise

```
q = 0.5
budget = 2
penalty = 2
```

```
portfolio = PortfolioOptimization(
    expected_returns=mu, covariances=sigma,
    risk_factor=q, budget=budget
)
```

```
qp = portfolio.to_quadratic_program()
```

```
algorithm_globals.random_seed = 1234
```

```
cobyla = COBYLA()
cobyla.set_options(maxiter=500)
ry = TwoLocal(num_assets, "ry", "cz", reps=3,
    entanglement="full")
svqe_mes = SamplingVQE(sampler=Sampler(), ansatz=ry,
    optimizer=cobyla)
```

Existing Approaches

- Quantum SDKs
 - Require programming and quantum expertise
- LLM-based code generation
 - No reproducibility or execution guarantees

```
q = 0.5  
budget = 2  
penalty = 2
```

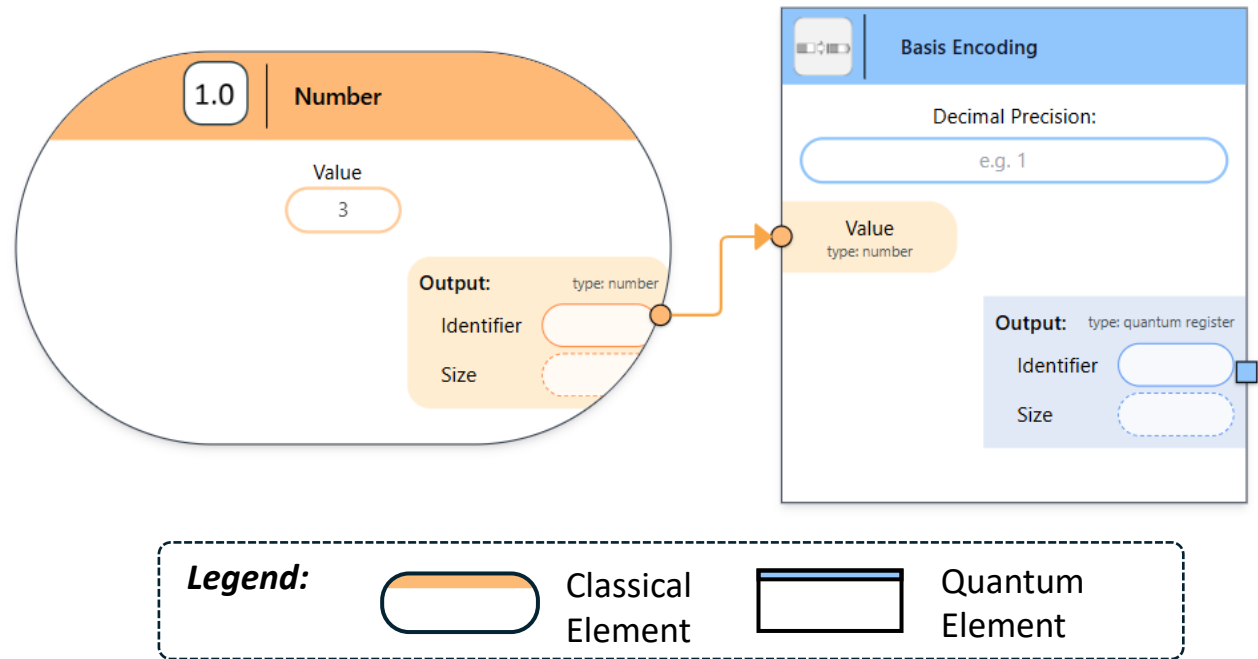
```
portfolio = PortfolioOptimization(  
    expected_returns=mu, covariances=sigma)  
qp = to_quadratic_program(portfolio)
```

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algorithm_globals.random_seed = 1234
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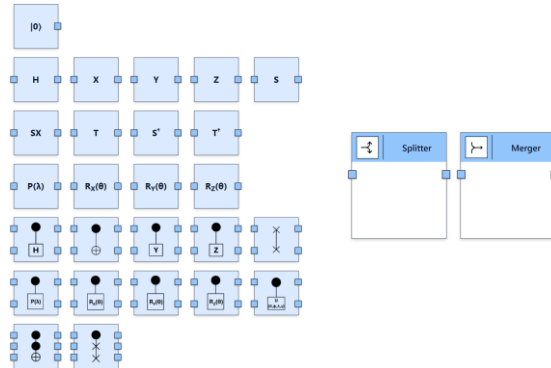
Existing Approaches

- Quantum SDKs
 - Require programming and quantum expertise
- LLM-based code generation
 - No reproducibility or execution guarantees
- Quantum low-code platforms
 - Domain-agnostic



Modeling Language

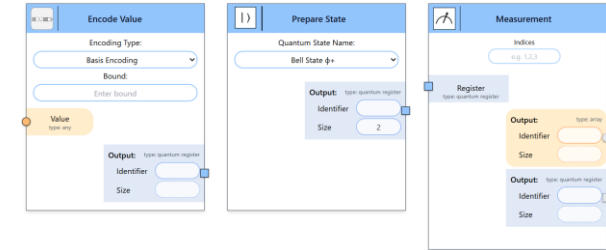
Circuit Primitives



Data Types



Boundary Elements

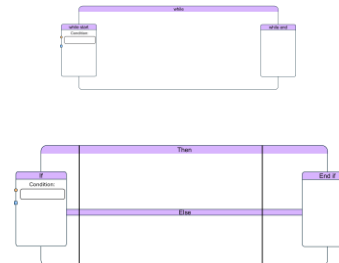


Modeling Elements

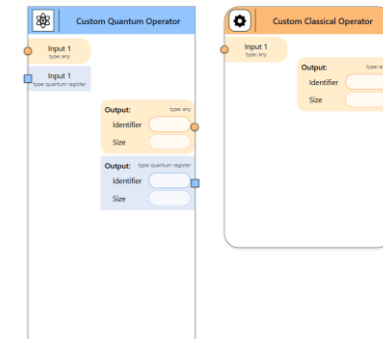


Operators

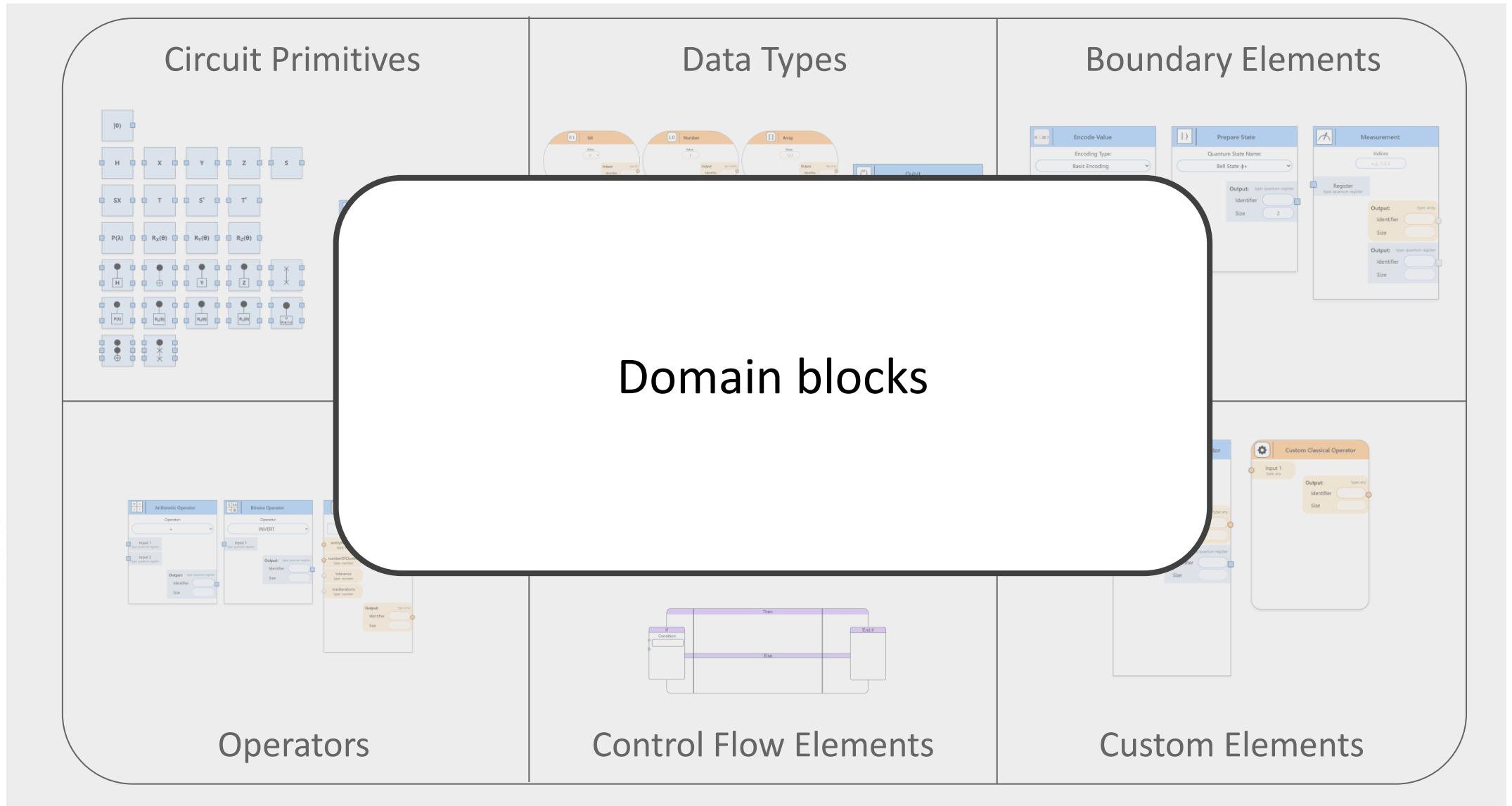
Control Flow Elements



Custom Elements



Modeling Language



Domain Blocks

- Represents a domain-relevant task
- Encapsulates underlying quantum or classical logic and exposes only domain-familiar inputs and outputs

Inputs

Domain-typed inputs

Aliases

Human-readable synonyms for discoverability

Metadata

Label, icon, description for the visual palette

Mapping

Connects domain block to one or more low-code block graphs

Outputs

Domain-typed outputs

Constraints

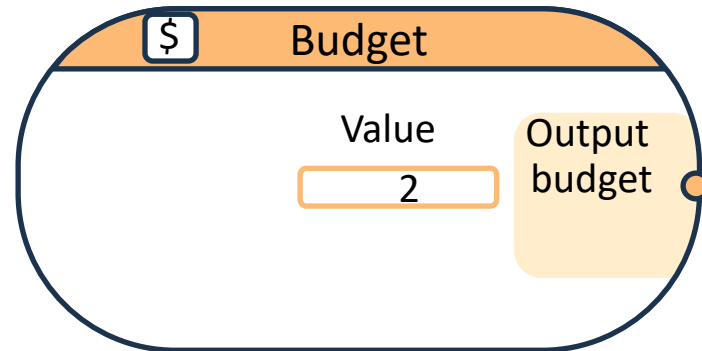
Domain rules

Defaults

Preset values

Domain Blocks

- Represents a domain-relevant task
- Encapsulates underlying quantum or classical logic and exposes only domain-familiar inputs and outputs



Domain Profile Phases

**Domain
Profile
Creation**

A

Legend:



Domain
Profile



Repository



Flow



Agentic
Subprocess



End
Event



Verified



Start
Event



Gateway



Task

Domain Profile Creation

Domain Blocks

Available to domain experts in the modeling palette

Constraints

Rules ensuring only valid compositions can be created

Defaults

Pre-filled values across blocks to reduce configuration effort

Templates

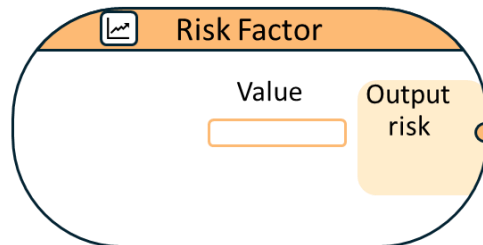
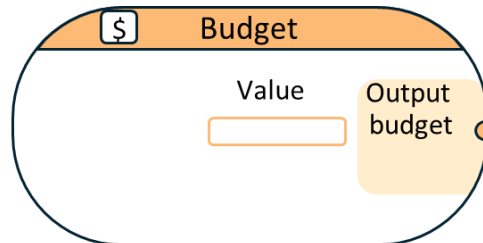
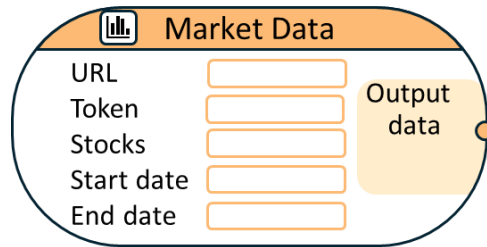
Ready-to-use model skeletons

Version

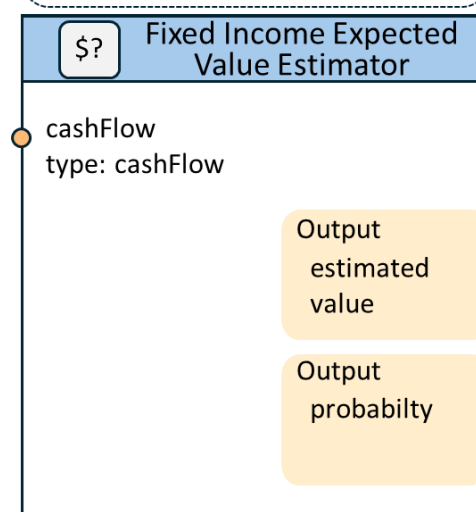
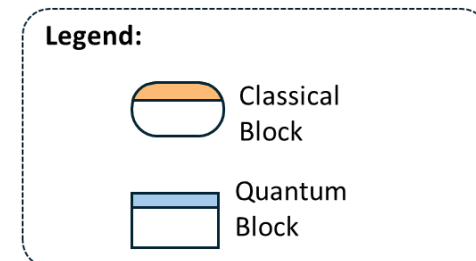
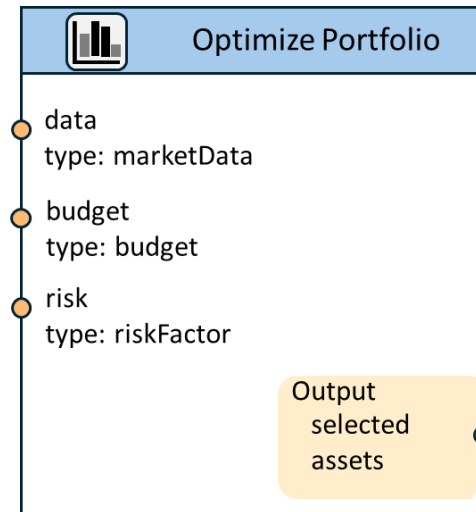
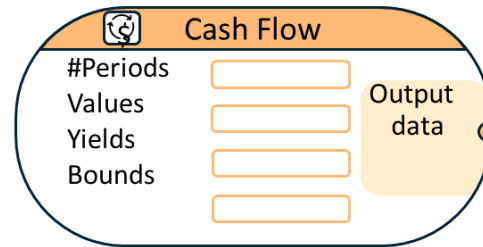
Each profile is versioned

Domain Profile Creation

Domain Blocks



Available to domain experts in the modeling palette



Domain Profile Creation

Domain Blocks

Available to domain experts in the modeling palette

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Rules ensuring only valid compositions can be created

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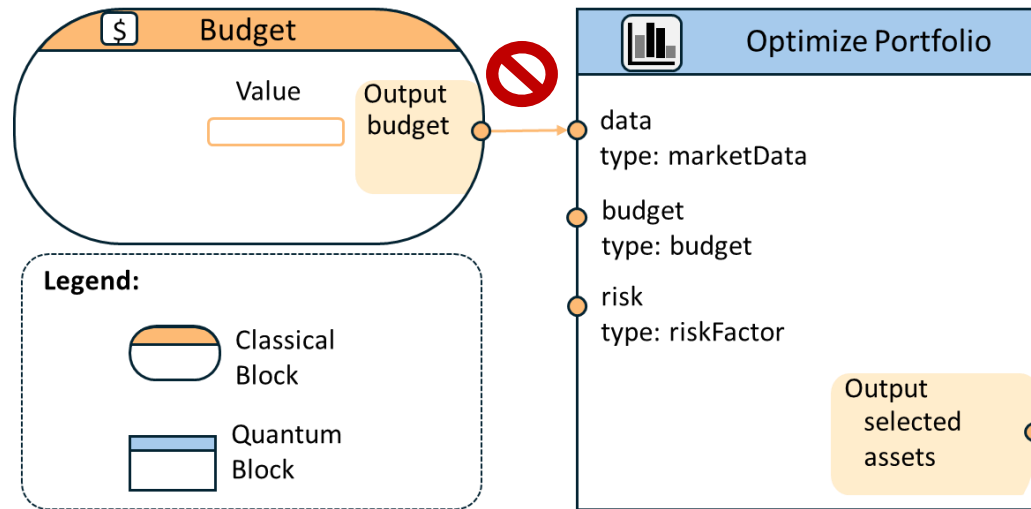
Domain Profile Creation

Domain Blocks

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Domain Profile Creation

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Domain Profile Creation

Domain Blocks

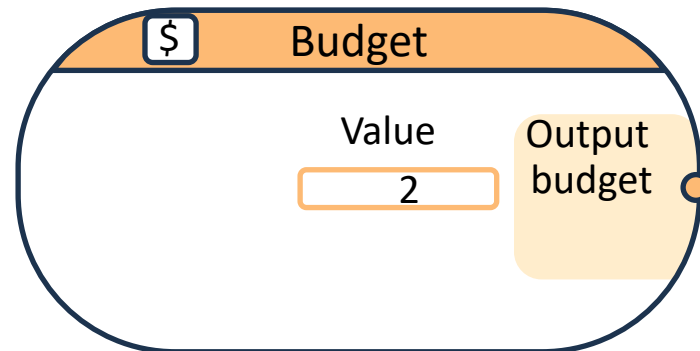
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Domain Profile Creation

Domain Blocks

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Domain Profile Phases

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Legend:



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Profile



Repository



Flow



Agentic
Subprocess



End
Event



Verified



Start
Event

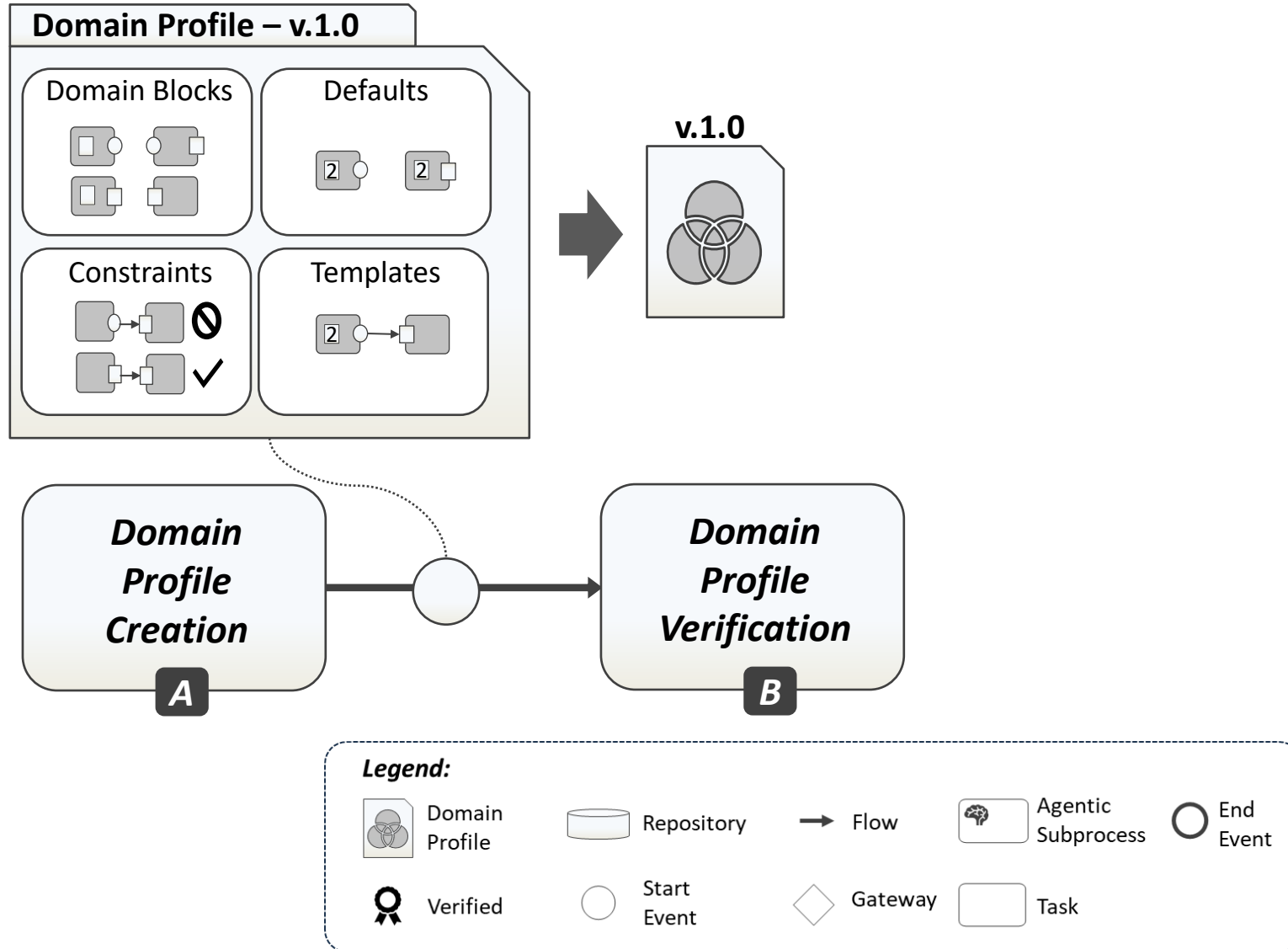


Gateway

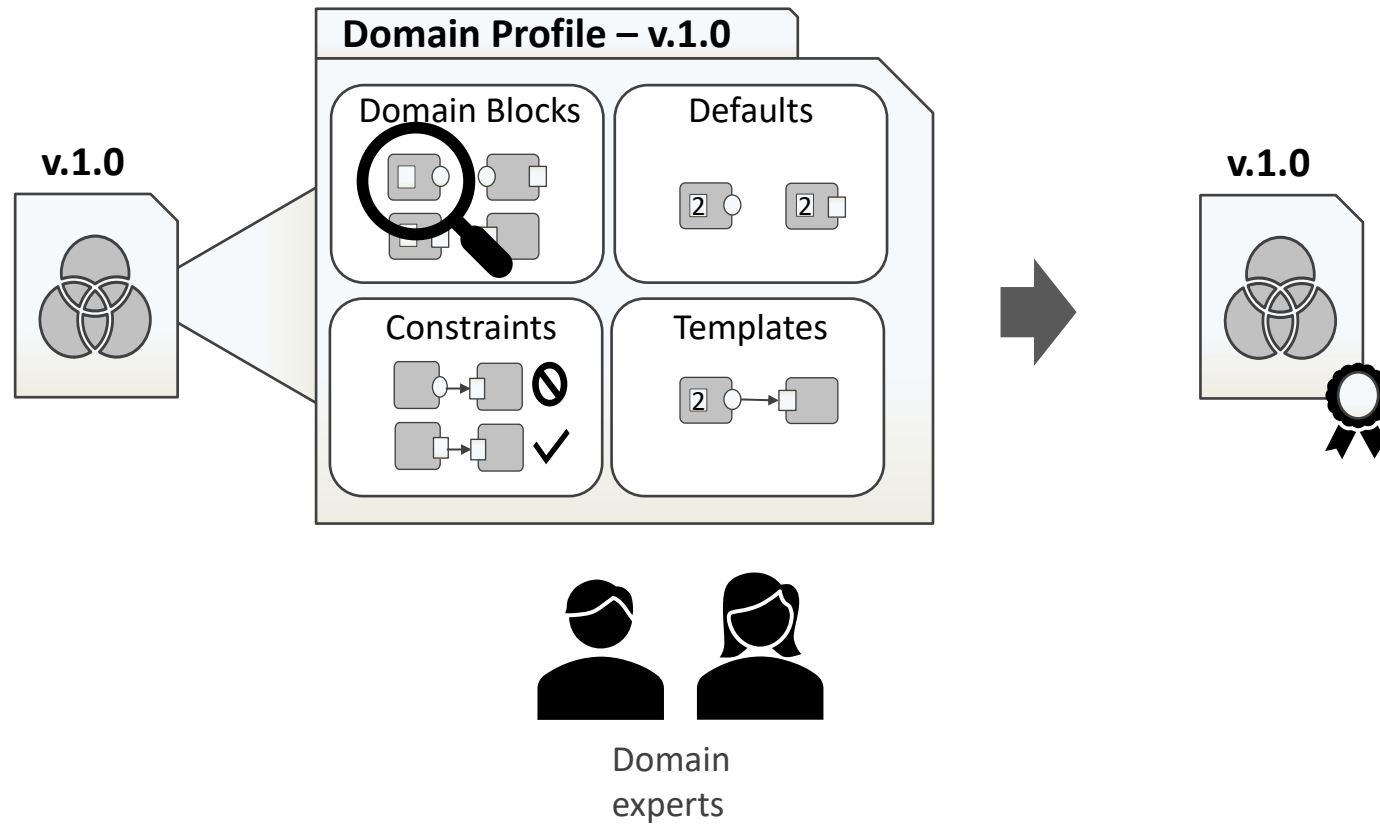


Task

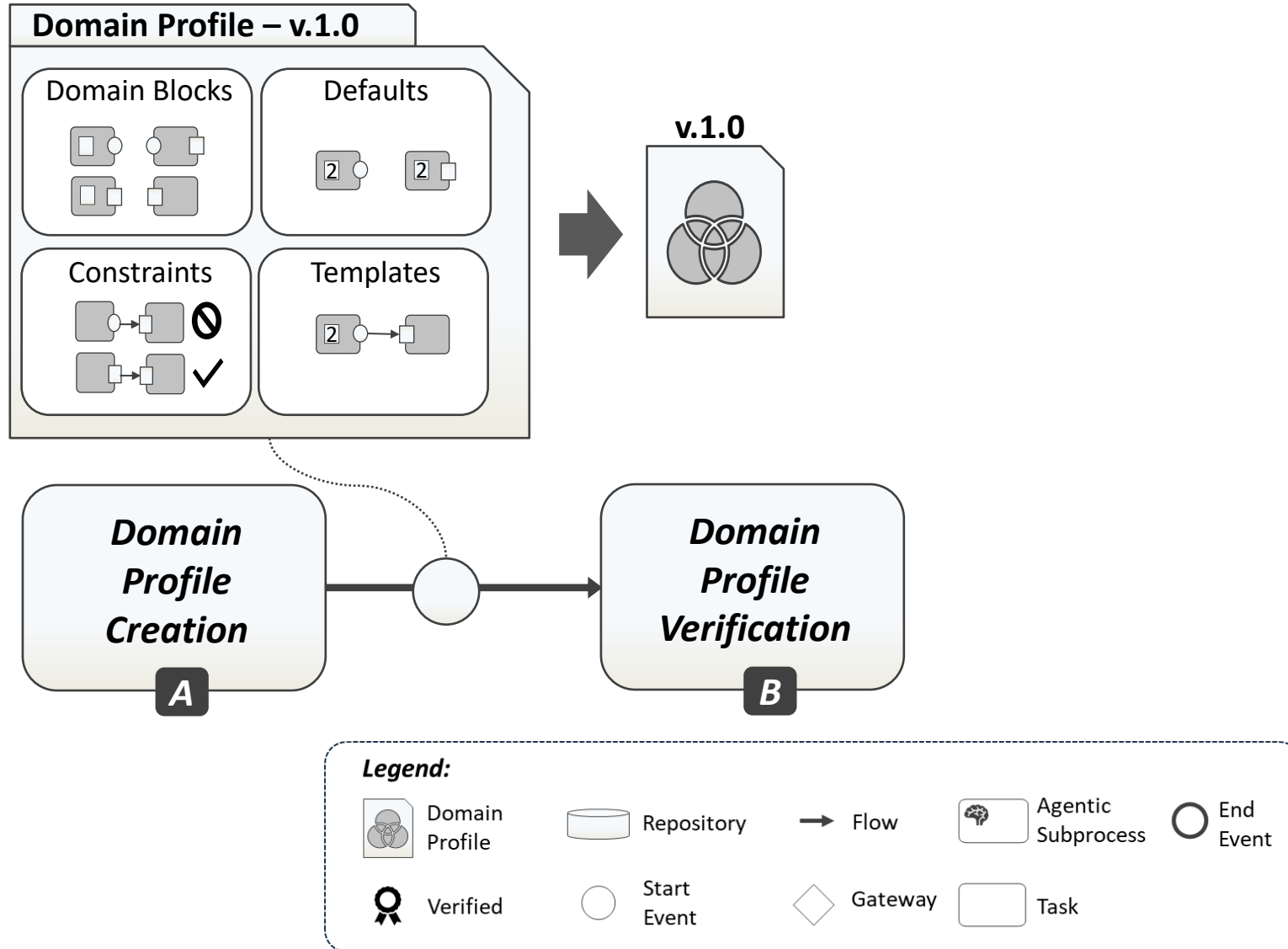
Domain Profile Phases



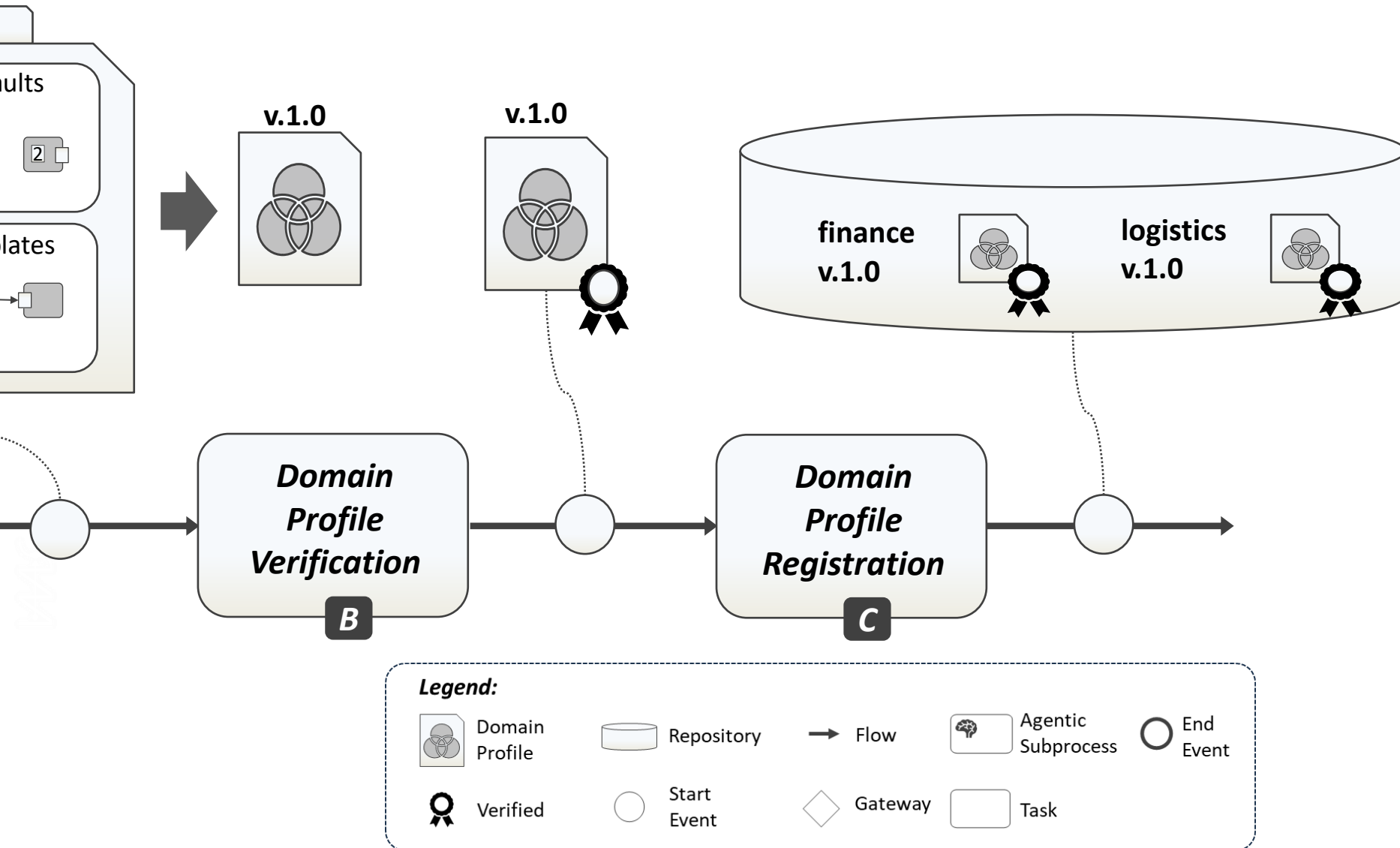
Domain Profile Verification



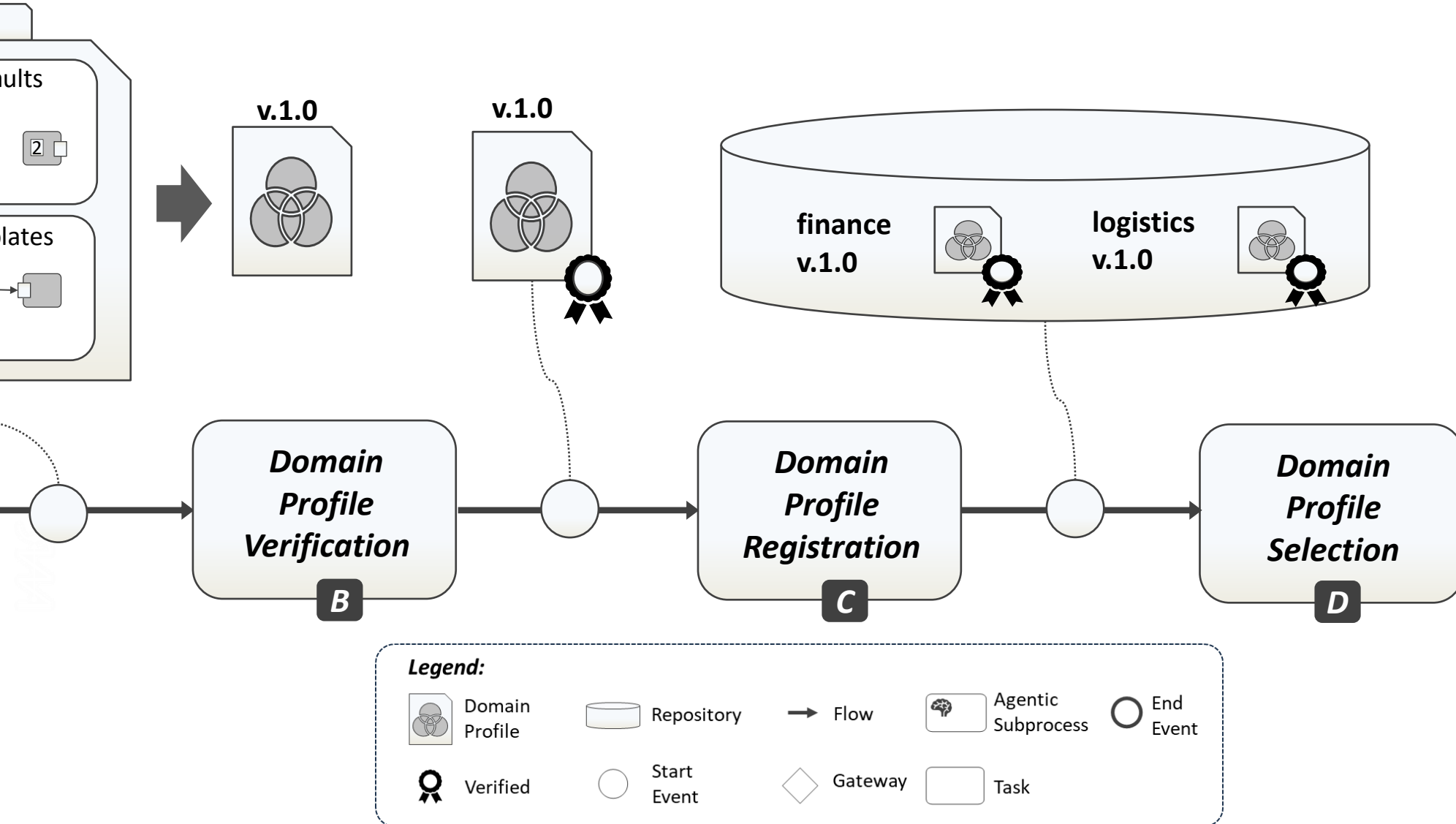
Domain Profile Phases



Domain Profile Phases



Domain Profile Phases



Domain Profile Selection

Qconstruct

Manage TemplatesConfigurationManage Domain ProfilesSend to BackendDetect AlgorithmHistoryHelp

Search nodes...

Boundary Elements

Blocks that convert between classical and quantum information.

Circuit Primitives

Basic blocks for quantum circuits.

[4,6]
1.0.3Data Types

Classical data types such as bits and arrays used to define block inputs.

+ < Operators

Predefined quantum operators.

Custom Operators

User-defined custom operation with arbitrarily classical and quantum inputs/outputs.

Machine Learning (ML) Elements

Elements for implementing classical and quantum machine learning algorithms focused on clustering, classification, and dimensionality reduction.

Templates

Configurable quantum algorithms directly to use.

Experience Mode

Domain Profile

None

Experience Level

None

Finance

Logistics

Ancilla Modeling

Compact Visualization

Off

Completion Guaranteed

Yes

Collaboration Mode

Off

Default Name

Recommended Settings

Experience Level	Ancilla Mode	Compact Visualization
Explorer	Off	Off
Pioneer	On	On

Model Information

Version

1.0.0

Name

My Model

Author

Description

This is a model.

Optimization

Enable Width Optimization

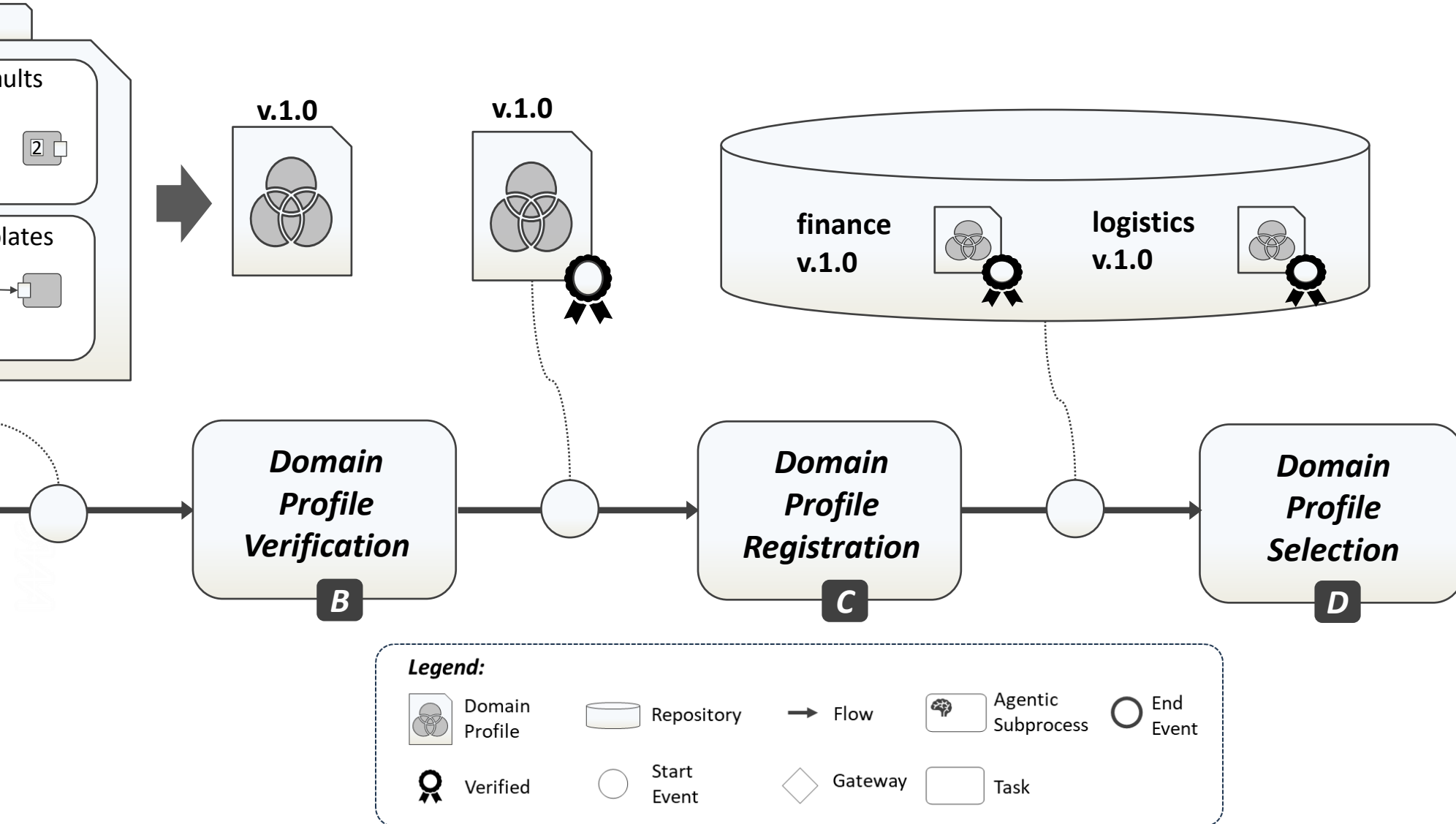
Enable Depth Optimization

React Flow

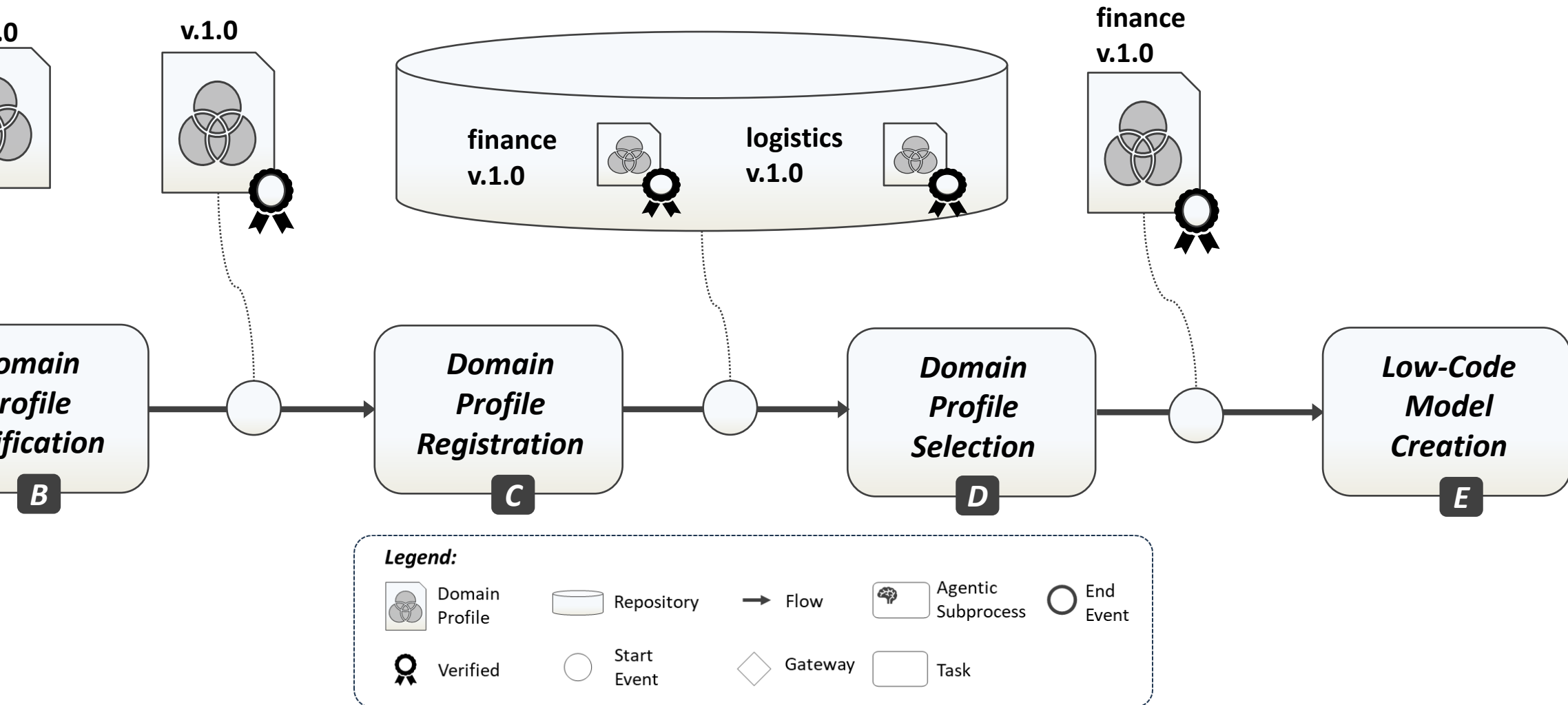
© Lavinia Stiliadou

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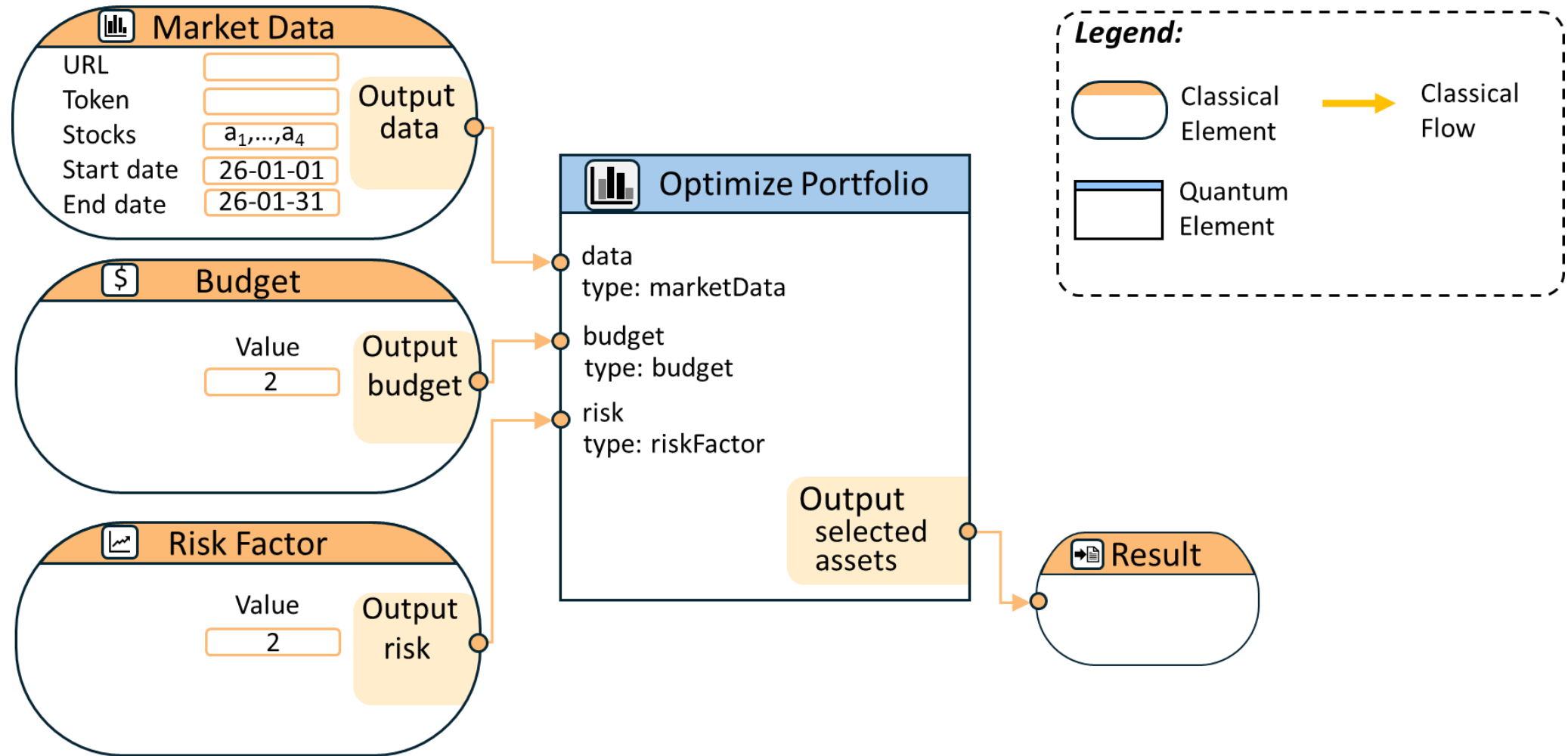
Domain Profile Phases



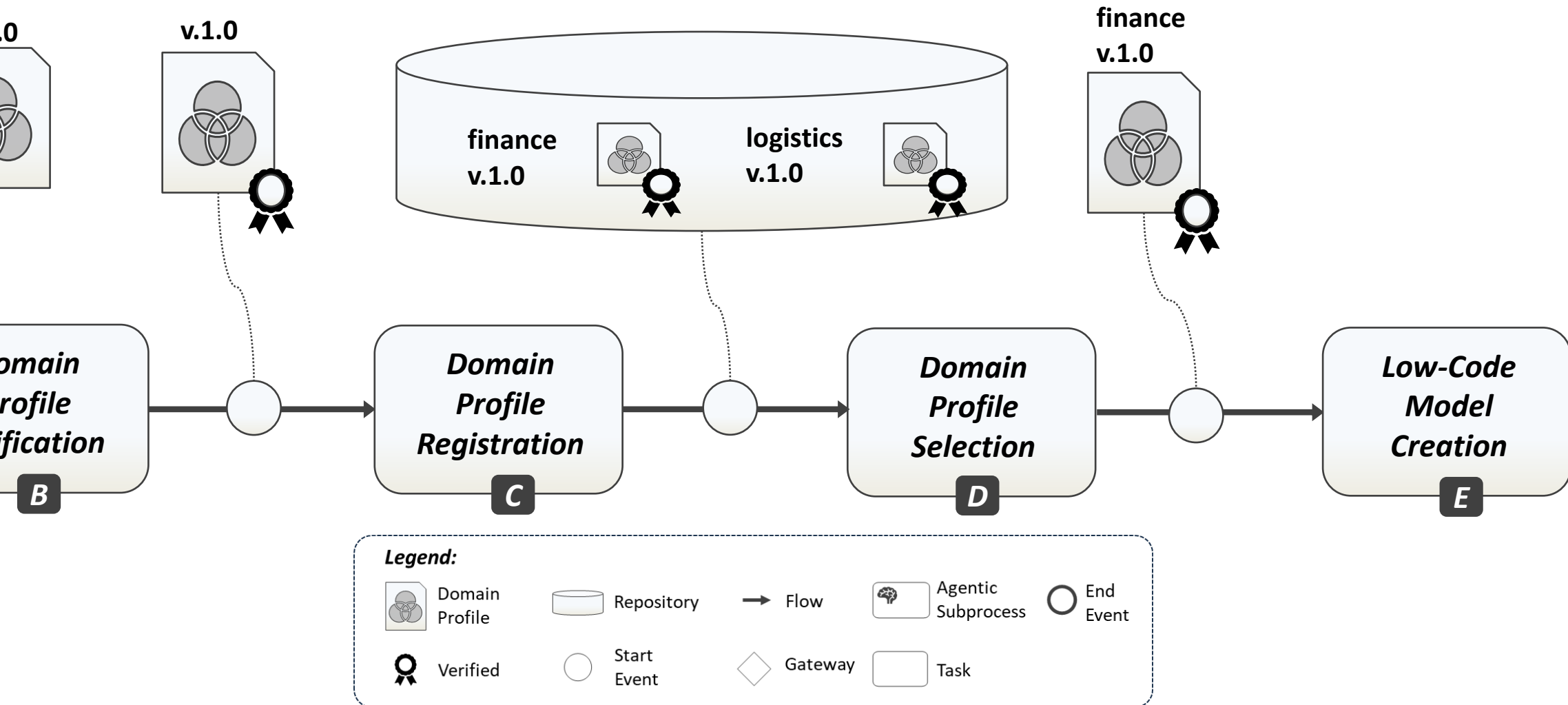
Domain Profile Phases



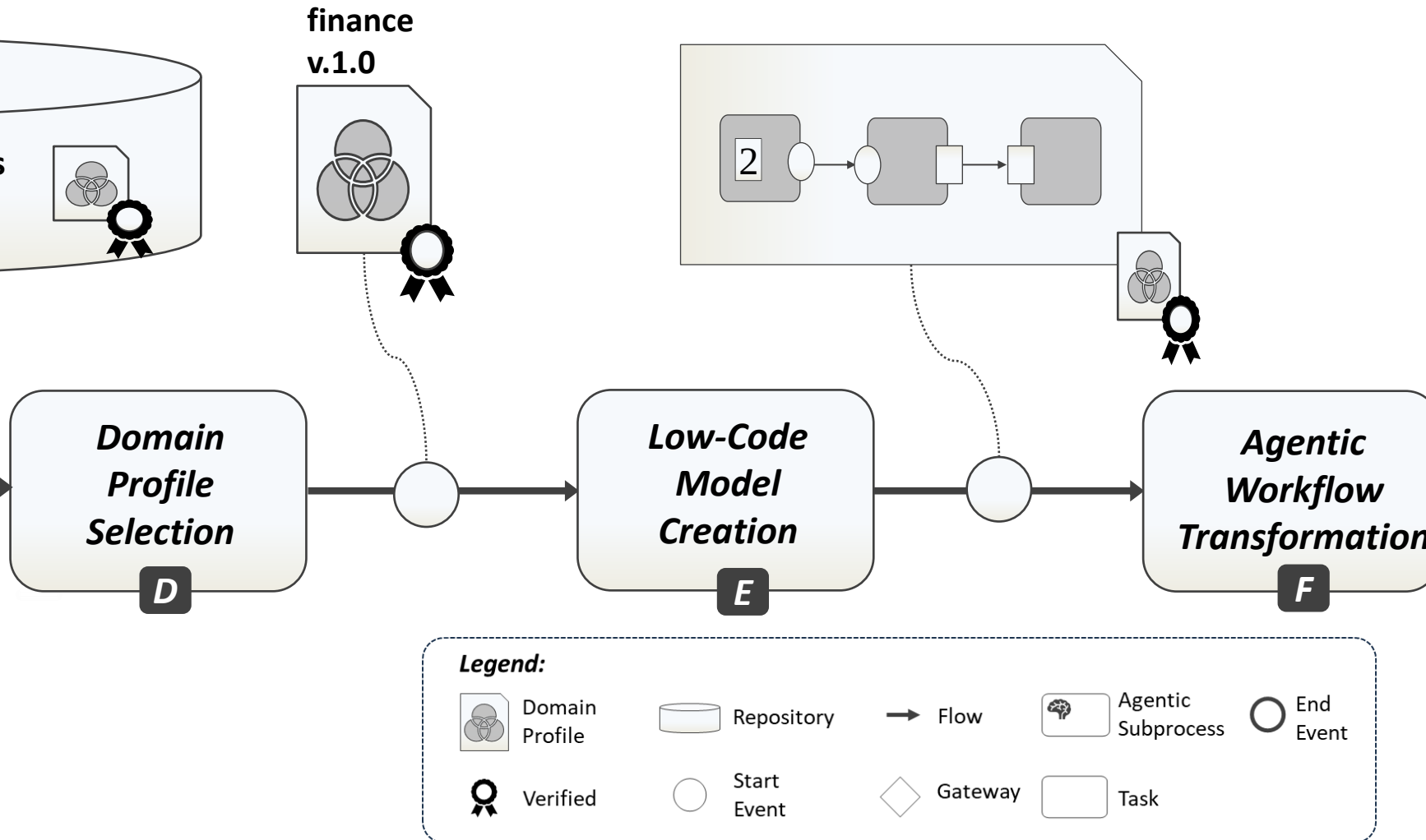
Low-Code Model Creation



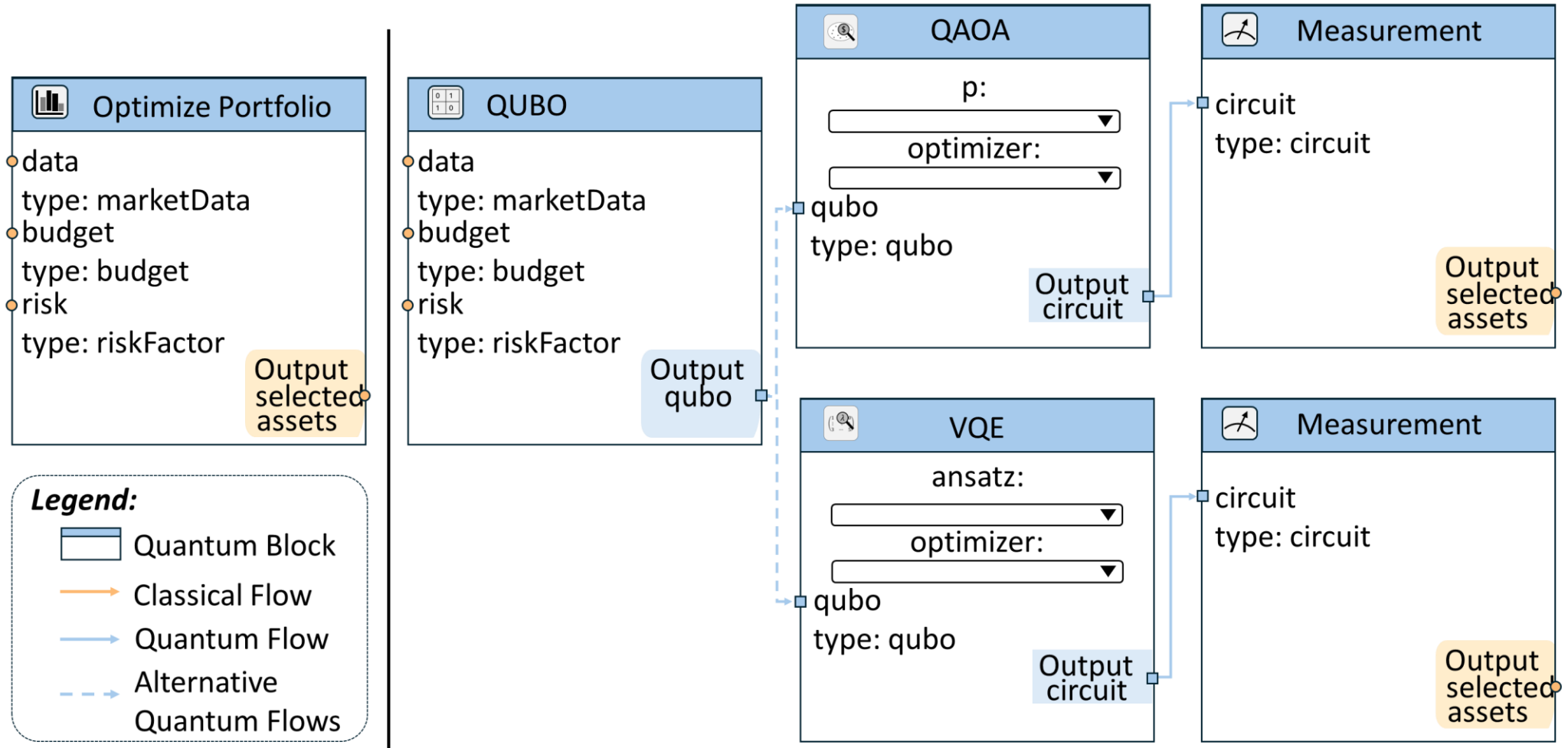
Domain Profile Phases



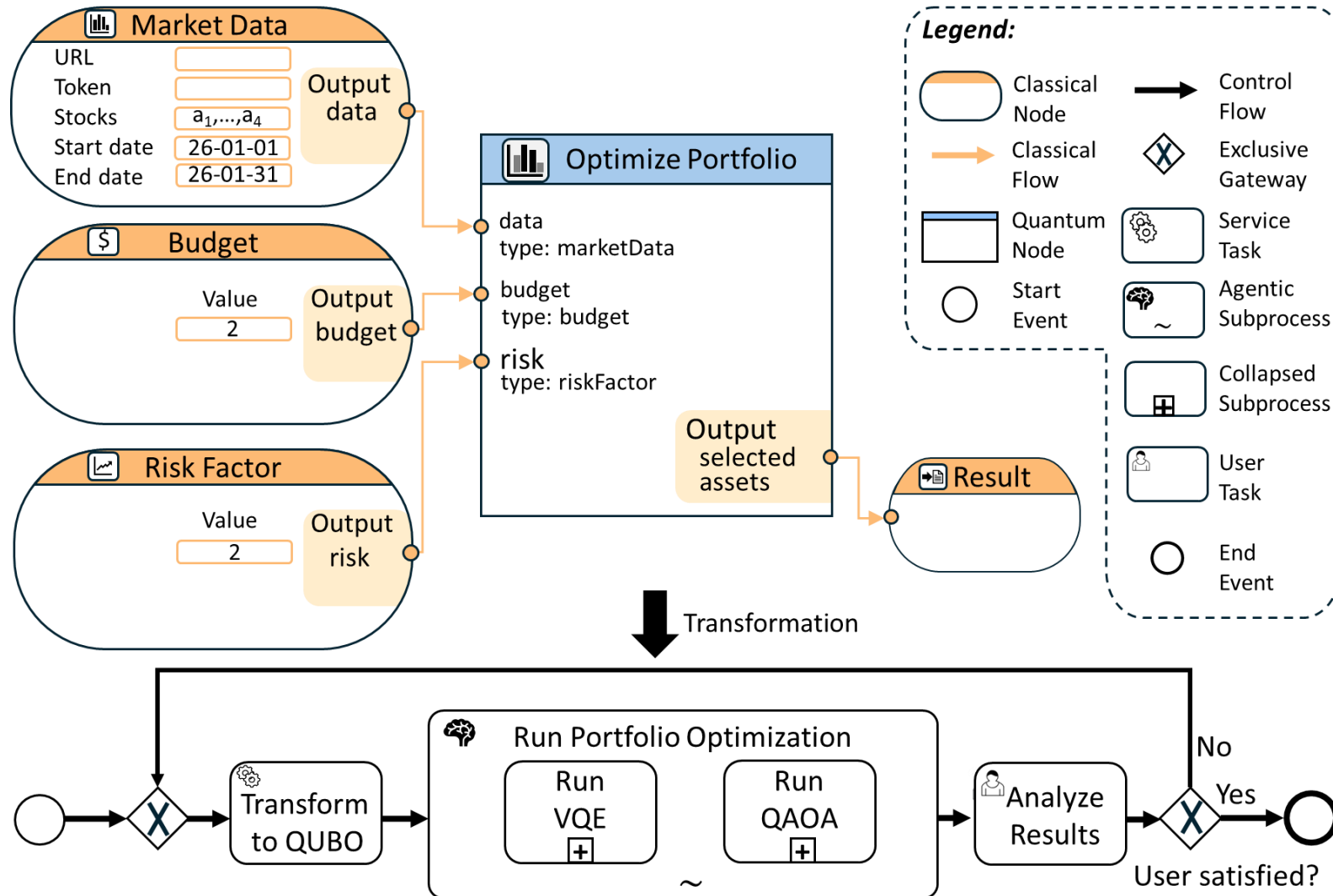
Domain Profile Phases



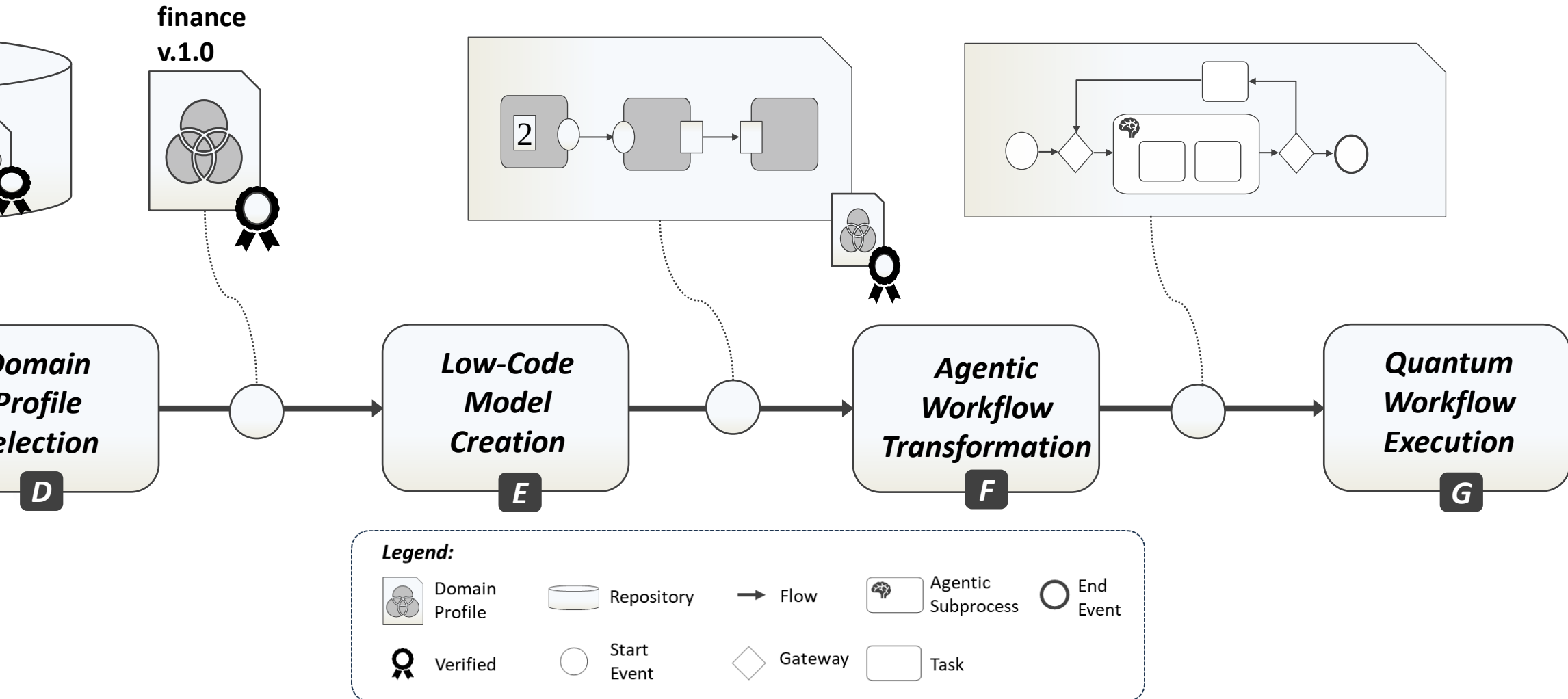
Domain Block Transformation



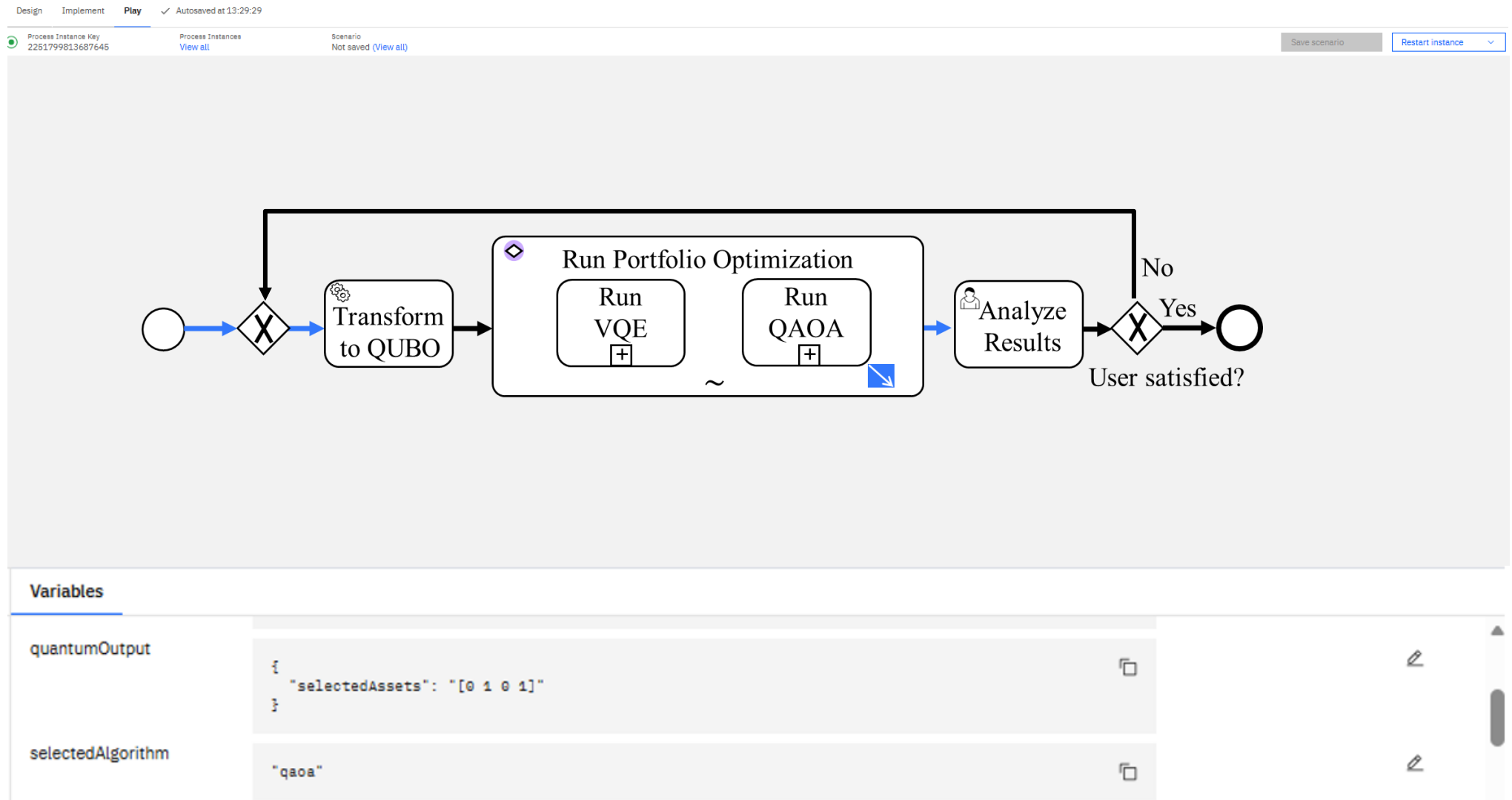
Agentic Workflow Transformation



Domain Profile Phases



Quantum Workflow Execution



Evaluation

Participants

- 8 domain experts total
- 4 finance experts
- 4 logistics experts
- Limited or no prior quantum programming experience

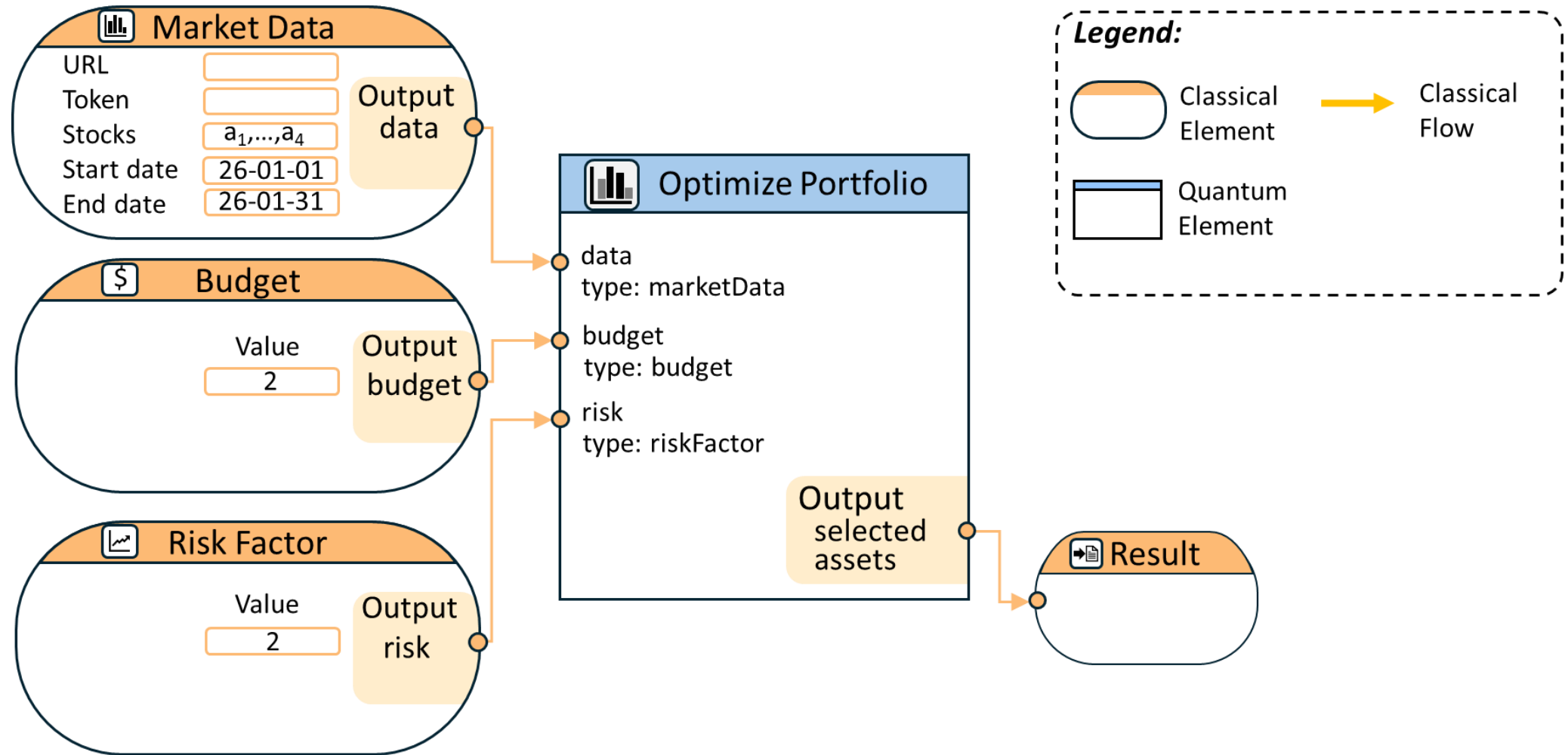
Study Design

- Within-subjects design
- Condition A: General-purpose low-code blocks (no domain profile)
- Condition B: Domain profile with high-level blocks
- 90 min per session (tutorial + modeling + questionnaire)

Collected Metrics

- Task completion time
- Number of blocks used
- Number of modeling steps
- Prevented invalid inputs
- Perceived ease of use (1–5)
- Clarity of concepts (1–5)
- Confidence in correctness (1–5)

Use Case 1



Evaluation – Finance Domain

- Use Case 1 (UC1):
 - Portfolio Optimization

<i>Metrics</i>	<i>UC1 (without profile)</i>	<i>UC1 (with profile)</i>
Task completion time (min)	30 / 35	10
Number of blocks	12 / 12	5
Prevented invalid inputs	0 / 0	2
Steps to complete model	32 / 32	16
Ease of use (1–5)	3 / 3	4
Clarity of concepts (1–5)	3 / 3	4
Confidence in correctness (1–5)	2 / 2	4

Note: slashes in 'no profile' column for UC1 indicate alternative algorithms (VQE/QAOA) that the user must manually choose without a domain profile.

Evaluation – Finance Domain

- Use Case 2 (UC2):
 - Income Estimation

<i>Metrics</i>	<i>UC2 (without profile)</i>	<i>UC2 (with profile)</i>
Task completion time (min)	25	12
Number of blocks	8	4
Prevented invalid inputs	0	4
Steps to complete model	19	11
Ease of use (1–5)	3	4
Clarity of concepts (1–5)	3	4
Confidence in correctness (1–5)	3	4

Results – Finance Domain Profile

<i>Metrics</i>	<i>UC1 (without profile)</i>	<i>UC2 (without profile)</i>	<i>UC1 (with profile)</i>	<i>UC2 (with profile)</i>
Task completion time (min)	30 / 35	25	10	12
Number of blocks	12 / 12	8	5	4
Prevented invalid inputs	0 / 0	0	2	4
Steps to complete model	32 / 32	19	16	11
Ease of use (1–5)	3 / 3	3	4	4
Clarity of concepts (1–5)	3 / 3	3	4	4
Confidence in correctness (1–5)	2 / 2	3	4	4

Note: slashes in 'no profile' column for UC1 indicate alternative algorithms (VQE/QAOA) that the user must manually choose without a domain profile.

➔ **Domain profile reduced task time from 30 min to 11 min, number of blocks from 10 to 5, and raised confidence in correctness from 2 to 4**

Evaluation – Logistics Domain

- Use Case 3 (UC3):
 - Vehicle Routing

<i>Metrics</i>	<i>UC3 (without profile)</i>	<i>UC3 (with profile)</i>
Task completion time (min)	18	12
Number of blocks	8	5
Prevented invalid inputs	0	2
Steps to complete model	20	14
Ease of use (1–5)	3	4
Clarity of concepts (1–5)	3	4
Confidence in correctness (1–5)	3	4

Limitations

- Profile creation effort
- Limited cross-domain reuse
- Scalability
- Agent quality

Conclusion

- Introduced a multi-domain quantum low-code platform to reduce fragmentation
- Combined:
 - Low-code modeling
 - Domain profiles
 - Agentic workflows
- Aims to reduce the entry barrier for non-experts in quantum software development



Thank you for your attention 😊

